

Recommender Systems

Collaborative Filtering

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Motivation

- Play a central role in how we interact with digital platforms today.
- Users face information overload
- Widely used in e-commerce, streaming, social media, etc.
- Many different kinds of Recommender Systems - collaborative filtering, content-based, context-aware, hybrids.
- Key challenge: Choosing the right model

Research Question

How do different collaborative filtering approaches compare in terms of recommendation accuracy, scalability, and robustness across different benchmark datasets?

- **Accuracy:** Which of the models gives the best recommendation accuracy on each dataset?
- **Scalability:** How do the models compare in training and prediction time across the datasets of different sizes?
- **Robustness:** How stable are the results of each model when tested on different types of datasets? Do the models perform differently on datasets with explicit ratings compared to implicit data?

Scope and Approach

Resource constraints, both in terms of available hardware and time.

- Collaborative filtering
- Ambitious enough to be meaningful, while still feasible to execute
 - 4 models
 - 4 datasets
- Explicit and Implicit data representation
- Unified experimental framework (RecBole)
- Same metrics & setup for all experiments

Models Overview

Models chosen for increasing complexity

- ItemKNN - Item-based K-Nearest Neighbor,
 - No training
 - Item-based nearest neighbor method that recommends items based on similarity to items the user has previously interacted with.
- BPR - Bayesian Personalized Ranking
 - Learns user and item embeddings by optimizing pairwise ranking preferences rather than predicting
- NeuMF - Neural Matrix Factorization
 - Extends classical matrix factorization using neural networks to model non-linear user-item interactions in parallel
- GCMC - Graph Convolutional Matrix Completion
 - Represents the recommendation problem as a bipartite graph, and predicts the value of unobserved entries using an auto-encoder followed by a pairwise (user-item) decoder model.

Datasets

Different characteristics in terms of size, sparsity, and rating structure.

- Amazon All Beauty
 - Extremely sparse and representative of real-world e-commerce data, where most users interact with only a very small number of items.
- ML-100K
 - classic recommender systems benchmark. Small and good reference point.
- Douban
 - Large number of users but only a small number of items.
- Jester
 - Dense and contains continuous ratings on a wide scale.

Dataset	#User	#Item	#Interaction	Sparsity	#Unique Ratings	Rating Range
amazon_all_beauty	631,986	112,565	693,929	99.9990%	5	[1, 5]
ml-100k	943	1,682	100,000	93.70%	5	[1, 5]
douban	738,701	28	2,125,056	89.73%	5	[1, 5]
jester	73,421	101	4,136,360	44.22%	412	[-10, 10]

Data Representation and Preprocessing

Representations

- expR: true explicit ratings
- impR: implicit interactions with rating values kept
- impB: binary implicit feedback

Preprocessing

- Remove one interaction for users who interacted with all items
- Remove duplicate user-item interactions (keep highest rating)
- Remove users with fewer than 5 interactions
- For implicit datasets - Remove interactions below rating threshold

Data Splitting Strategy

- Leave-one-out per user
 - Seemed widely used.
- Uniform random sampling
 - For comparability (not all datasets contain temporal data)
- Same strategy for all datasets

Evaluation Metrics

Rating vs Ranking-based

Standard cutoffs reflecting realistic recommendation list sizes - $K = 10$ and 20

- **Precision@K** Fraction of the top- K recommended items that are relevant
- **Recall@K** Fraction of relevant items that appear in the top- K recommendations
- **MRR@K** (Mean Reciprocal Rank) How early the first relevant item appears in the ranked list
- **Hit@K** (Hit Rate) Whether at least one relevant item appears in the top- K
- **NDCG@K** (Normalized Discounted Cumulative Gain) Measures ranking quality, rewarding relevant items appearing earlier

Effectively only 4 Metrics as $Recall@K \equiv Hit@K$ for Leave-one-out

Experimental Setup

Experimental framework **RecBole**:

- Same setup used across all models and datasets
- No hyperparameter tuning performed
- Default model configurations used
- Leave-one-out evaluation with random sampling
- Ranking-based metrics at $K = 10$ and 20
- Early stopping
 - NDCG@10
 - Five consecutive epochs without improvement
 - Max 50 epoch
- Training eval - Uniform negative sampling with 100 negative items per positive.
- Final test - Full ranking.

Accuracy Results

No single model performs best across all datasets

- **GCMC**: Best overall accuracy, especially on explicit datasets
- **NeuMF**: Best performance on ML-100K
- **ItemKNN**: Best on Douban
- **BPR**: Generally not good or bad accuracy

Efficiency & Scalability

More pronounced differences between models

- **ItemKNN**
 - Fastest
 - No training, slightly higher eval time.
 - Minimal resources.
 - Well suited for frequently updated systems.
- **BPR & NeuMF**
 - Moderate training and evaluation cost
 - NeuMF has slightly higher training time due to neural component
- **GCMC**
 - Highest computational cost
 - GPU-intensive - 24H vs 3H.
 - Clear tradeoff between accuracy and computational cost.

Robustness Analysis

- **impR $\approx$ impB**
 - Nearly identical performance
 - Retaining rating values adds little in implicit settings
- Models show consistent *relative* behavior across datasets
 - **GCMC** generally most accurate.
 - **NeuMF** competitive on ML-100K.
 - **BPR** somewhere in the middle.
 - Generally similar performance with the exception of **ItemKNN** on Jester (may be do to continuous ratings).

Conclusion & Takeaways

No universally best collaborative filtering model

- Dataset characteristics matter most
- Recommendation task matters.
- Graph models give best accuracy (at high cost)
- Simpler models are often sufficient and more practical
- Model selection should be application-specific (accuracy, scalability, robustness)

Note on Model Selection

Not all models naively support both explicit and implicit feedback, limiting direct comparability between explicit and implicit results.

- **ItemKNN** — similarity-based, originally implicit
 - Easy fix - Load Real values in interaction matrix
- **BPR** — pairwise ranking, implicit feedback only
 - Modify the loss so higher ratings matter more (Weighted BPR)
 - $\mathcal{L} = - \sum_{(u,i,j) \in D} w_{u,i} \log \sigma(\hat{y}_{u,i,j})$ Where $w_{u,i}$ is some kind of weight term
- **NeuMF** — neural MF, implicit feedback (binary loss)
 - Use Multiclass Cross-Entropy instead of Binary Cross-Entropy
 - With softmax prediction output like GCMC
- **GCMC** — graph neural network, supports explicit ratings

Note on Data Representation

Different prediction tasks with opposing effects on top-K ranking metrics.

- expR - **1** relevance representation
 - Did user interact with item?
 - Potentially weak signal $\rightarrow$ harder ranking
 - Less negative items $\rightarrow$ easier ranking.
- impR and impB **2** relevance representations
 - Did user like the item?
 - Stronger signal $\rightarrow$ easier ranking
 - More negative items $\rightarrow$ harder ranking.

Dataset variant	Task	Feedback type
expR	Interaction prediction	Explicit
<i>expR $\rightarrow$ binary</i>	<i>Interaction prediction</i>	<i>Implicit</i>
impR	Preference prediction	Implicit (weighted)
impB	Preference prediction	Implicit

Table: Categorization of dataset variants by task and feedback type.

Final remarks